PROJECT DESCRIPTION:

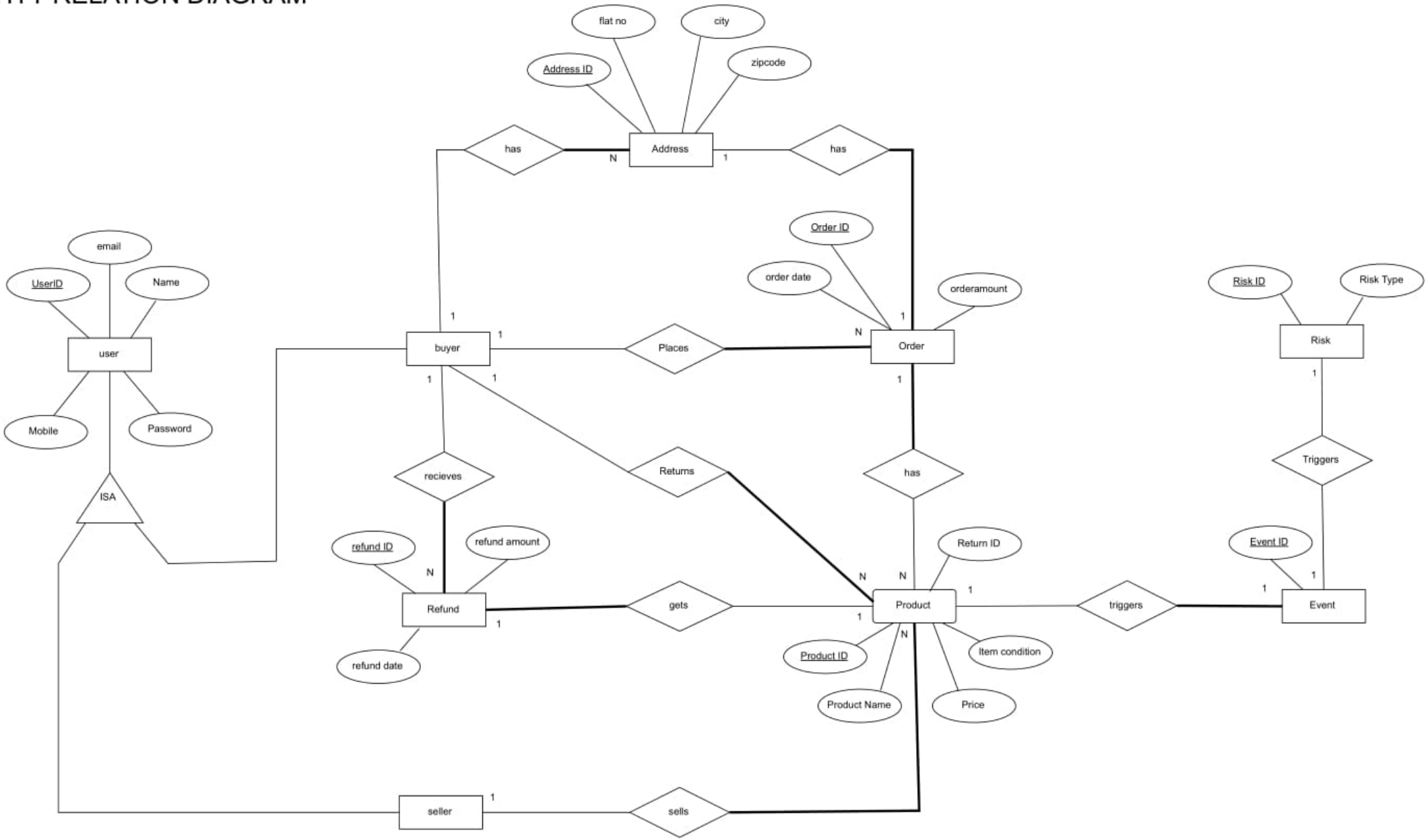
The main objective of this project is to develop an E-Commerce database and to identify one of the risks, a site can face from the customers who in this case are Buyers. Buyers are essential to keep the site function at all times. Every buyer has a choice to return a product, if received in different condition than ordered. The more the number of returns, the less is the profit. The Admin of the E-commerce monitors delivery of the products, refunds and returns related to the Buyer. Analysing these reports Admin manages the E-commerce for better service, profitable and smooth operations.

REQUIREMENT ANALYSIS:

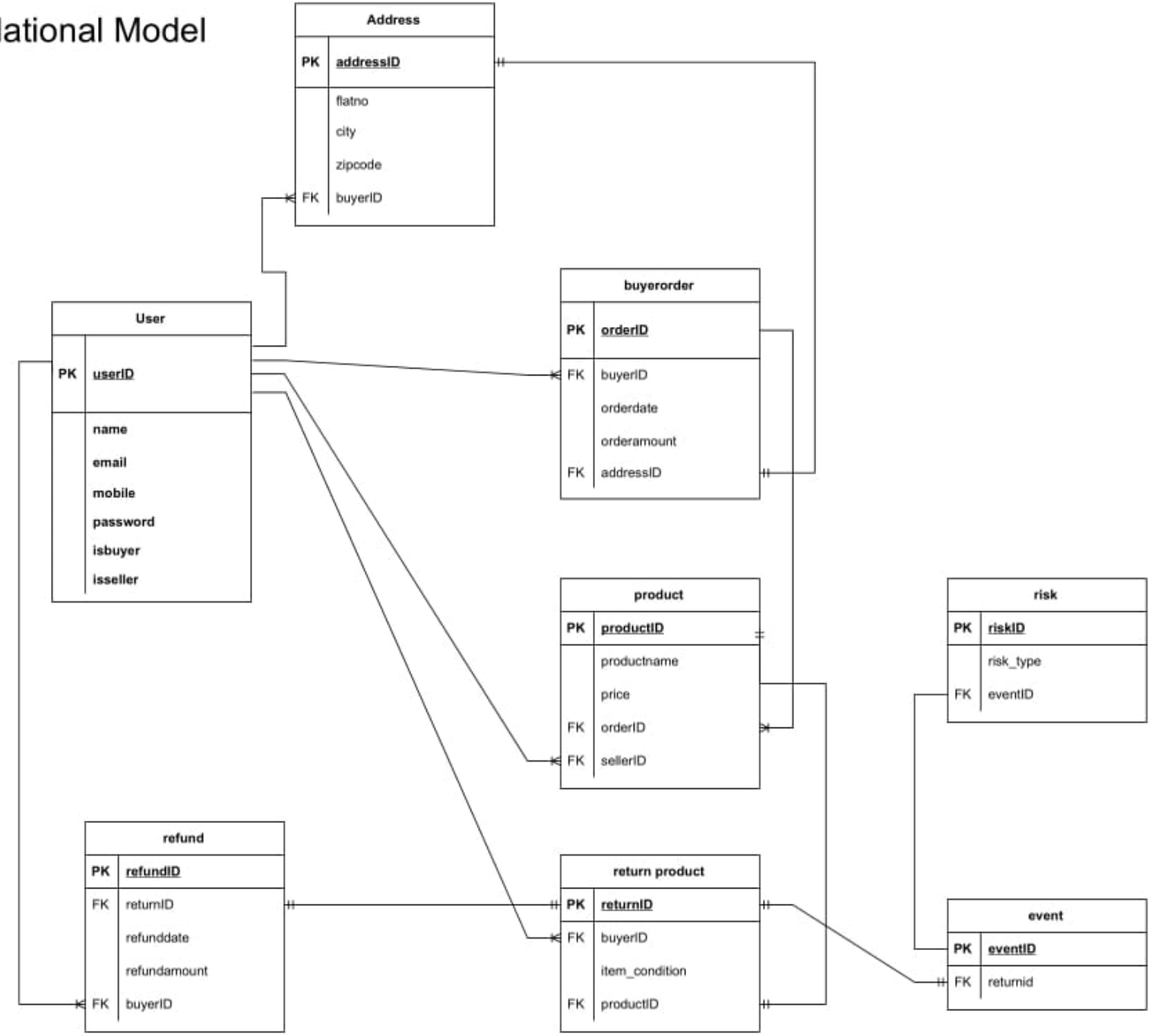
The database to be created would have the following: Entities (Red), Attributes (Blue) and Relationship (Green):

- Each User has one account with his basic detail like User ID, Name, Mobile, E-mail and Password.
- A User can be a Buyer or a Supplier or both.
- A Buyer can have multiple Address.
- Address will have Address ID, Flat number, City and Zipcode.
- A Buyer can place multiple Orders.
- Each Order will have Order ID, Order date and Order amount.
- Each Order will have Address details.
- Order can have multiple Products.
- Each Product will have Product ID, Product Name, and Price.
- A Seller can sell multiple Products.
- A Buyer can return products for Refund.
- Every Refund has a Refund ID, Refund amount and Refund date.
- Every returned Product will have a Return ID and Item condition.
- All the Returns with bad item condition are lodged as an Event.
- Each Event will have an Event ID.
- An Event with a condition is project the Risk.
- Each Risk has a Risk ID and Risk type.

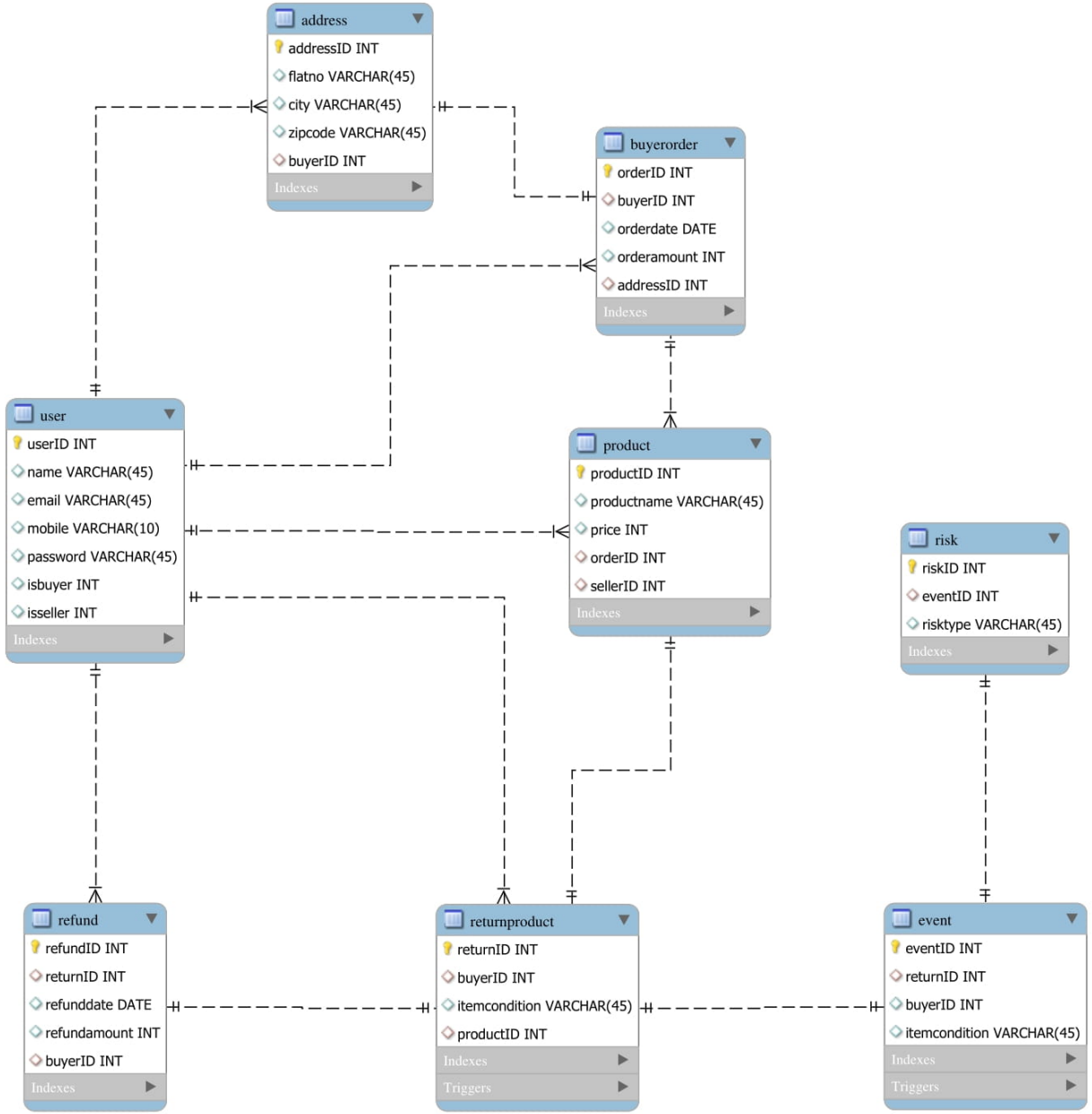
ENTITY-RELATION DIAGRAM



Transformation to Relational Model



MySQL Tables



SQL Queries

1. List all the users who have returned products with bad item condition from region zipcode E1F1.

Query:

```
select u.name, p.productname, r.itemcondition, a.zipcode
from user u
     join returnproduct r on u.userID=r.buyerID
     join product p on p.productID=r.productID
     join address a on a.buyerID=u.userID
where r.itemcondition like 'bad' and a.zipcode='E1F1'
```

Result:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name	productname	itemcondition	zipcode
Krupaul	Asus g15	bad	E1F1
Krupaul	Apple Macbook	bad	E1F1
Animon	Lenova ideapad	bad	E1F1
Animon	Apple Macbook	bad	E1F1
Animon	Asus g15	bad	E1F1

2. List the product details which availed refund in the months from June 2022 to August 2022.

Query:

```
select p.productname, f.refunddate
from product p
   join returnproduct r on p.productID=r.productID
   join refund f on f.returnID=r.returnID
where f.refunddate between '2022-06-01' and '2022-08-31'
```

Result:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

productname	refunddate
Asus g15	2022-06-06
Lenova ideapad	2022-07-07
Asus g15	2022-08-08

3. List the users who have returned 3 products or more in bad item condition and assign risk type as high risk.

Query:

```
select r.buyerID, u.name, i.risktype
from event e join risk i on i.eventID=e.eventID
   join returnproduct r on r.returnID=e.returnID
   join user u on r.buyerID=u.userID
   group by r.buyerID having count(r.buyerID)>=3
```

Result:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

buyerID	name	risktype
1003	Animon	high risk

Syntax for Trigger function for inserting tuples from return product with bad item condition to an event

```
CREATE TRIGGER `returnproduct_AFTER_INSERT` AFTER INSERT ON
`returnproduct` FOR EACH ROW
BEGIN
if new.itemcondition='bad' then
insert into event(eventID, returnID,buyerID,`itemcondition`)
values (eventID, new.returnID, new.buyerID,'bad');
end if;
END
```

Relational Algebra

1. List all the users who have returned products with bad item condition from region, zip code E1F1.

   $\Pi_{\text{name, productname, itemcondition, zipcode}}((\sigma_{\text{item Condition = 'Bad'}} \text{Return Product}) \bowtie \text{Product} \bowtie \text{user} \bowtie \text{Address})$

2. List the product details which availed refund in the months from June 2022 to August 2022.

   $\Pi_{\text{Productname, Refunddate}}((\sigma_{(\text{Refunddate} \geq 2022/06/01) \cap (\text{Refunddate} \leq 2022/08/31)} \text{Refund}) \bowtie \text{Return Product} \bowtie \text{Product})$

3. List the users who have returned 3 products or more in bad item condition and assign risk types as high risk.

   $\Pi_{\text{BuyerID, name, risktype}}(\text{Event} \bowtie \text{Risk} \bowtie \text{Return Product} \bowtie \text{user})$

   We are using an Aggregate operator Count in this Query.

Relational Calculus

1. List all the users who have returned products with bad item condition from region, zip code E1F1.

   {P| ∃R ∈ Return Product ∃D ∈Product ∃U ∈User ∃A ∈Address (R. item Condition = 'bad'∧ A. Zipcode= 'E1F1' ∧ R. BuyerID=U. UserID ∧ R. Product ID = D. Product ID ∧ U.UserID= A.BuyerID ∧ P.Name = U.Name ∧ P. Product name = D. Product name ∧ P. itemcondition= R.itemcondition ∧ P. Zipcode=A. Zipcode) }

2. List the product details which availed refund in the months from June 2022 to August 2022.

   {P | ∃F ∈Refund ∃D ∈Product ∃R ∈Return Product (F. refunddate ≥2022/06/01 ∧ F.refunddate ≤2022/08/31 ∧ F. ReturnID = R. ReturnID ∧ R. ProductID = D. ProductID ∧ P. Product name = D. Product name ∧ P. Refund date = F. Refund date) }

3. List the users who have returned 3 products or more in bad item condition and assign risk types as high risk.

   We are using an Aggregate operator Count in this Query.